# Norm-resolvent convergence to the non-compact Lorentzian Krein
spectral triple on $S^3 \times \mathbb{R}_t$ and the two-rate hybrid limit

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**Abstract**

We close the de-compactification gap (G2) of the Lorentzian propinquity program [29, 30] at the norm-resolvent / spectral level. Combining the parametric stability of Paper 45 [29] (and its strong-form extension Paper 46 [30]) with a standard compact-to-non-compact resolvent comparison, we establish a two-rate hybrid convergence

$$\mathcal{T}^L_{n_{\max},N_t,T(N_t)} \xrightarrow{\Lambda^L\text{-rate } O(\log n_{\max}/n_{\max}+T(N_t)/N_t)} \mathcal{T}^L_{S^3\times S^1_T(N_t)} \xrightarrow{\text{nr-rate } O(e^{-|\mathrm{Im}\, z|\, T(N_t)/2})} \mathcal{T}^L_{S^3\times\mathbb{R}_t}$$

along any admissible scaling $T(N_t) \to \infty$ with $T(N_t)/N_t \to 0$. The inner arrow is propinquity convergence (Paper 45 main theorem verbatim, at coupled cells; descoped — the inner arrow and the § 8 G2-metric closure are open pending repair of the upstream Paper 45 seminorm; see the Status note); the outer arrow is norm-resolvent convergence of the Lorentzian Dirac on the compactified spacetime $S^3 \times S^1_T$ to the Lorentzian Dirac on the genuine spacetime $S^3 \times \mathbb{R}_t$, with an explicit exponential tail bound $O(e^{-|\mathrm{Im}\, z|\, T/2})$ on compact-temporal-support subspaces, and is unaffected by the descope. The construction preserves the BBB [2] Krein fundamental symmetry, the Krein-positive cone $\mathcal{K}^+$, and the Krein-self-adjointness of the Lorentzian Dirac across the entire two-rate diagram.

*Honest scope.* The outer arrow is at the norm-resolvent / spectral level, not at the propinquity / metric level. Propinquity-level convergence to the genuine non-compact Lorentzian Krein triple would require a non-compact extension of Latrémolière's propinquity [14, 15], which is not in the published literature as of May 2026. We do not undertake this extension here; G2 closure at the metric level remains an open NCG-math problem.

*Three-carrier identification.* The norm-resolvent limit identifies three nominally distinct Krein-Lorentzian carriers as approximation schemes for the same physical operator: the periodic compact $S^3 \times S^1_T$ (Paper 45 substrate), the bounded Dirichlet interval $S^3 \times [-T_{\max}, T_{\max}]$ (Paper 43 [27] substrate), and the non-compact $S^3 \times \mathbb{R}_t$ (this paper's limit object). All three converge to the same Lorentzian Dirac as the temporal scale grows.

*Place in the math.OA standalone series.* Ninth in the GeoVac math.OA arc (siblings: Papers 38 [23], 39 [24], 40 [25], 42 [26], 43 [27], 44 [28], 45 [29], 46 [30]). The result composes Paper 45/46 verbatim on the inner arrow with standard Reed–Simon [18] § XIII.16 / Hekkelman–McDonald 2024 [11] compact-to-non-compact-domain machinery on the outer arrow.

## Status and scope (June 2026)

Three issues are established upstream of this paper: (i) the $\mathcal{K}^+$-restricted compressed-Dirac Lipschitz seminorm underlying $\Lambda^L$ vanishes identically on the truncated operator system —

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Krein-self-adjointness of the anti-Hermitian spatial part $i\,D_{\mathrm{GV}} \otimes I$ forces $\{J_L, D_{\mathrm{GV}} \otimes I\} = 0$, so the $\mathcal{K}^+$ compression annihilates the spatial Dirac, and the momentum-diagonal temporal multipliers commute with $\partial_t$ (verified bit-exact at $(n_{\max}, N_t) \in \{(2,3),(3,5)\}$); (ii) the "Latrémolière Thm 5.5 / Def. 3.4 / Def. 3.5" references cited for the tunneling-pair bound do not exist in the cited source — the device actually used is van Suijlekom's state-space Gromov–Hausdorff framework; (iii) the constant $2/(n_{\max}+1)$ is the maximum of the Plancherel mass distribution, not a cb-norm of any map used in the proofs (the smoothing map is unital completely positive, with cb-norm 1).

Consequently the *inner arrow* (Theorem 3.1, propinquity at coupled cells), which consumes Paper 45's $\Lambda^L$ quantity, is descoped, and the G2-metric closure claims of § 8 and § 9 (Theorem 8.3, Q1, Q2) rest on the same degenerate metric quantity and are descoped. The *outer arrow* — norm-resolvent convergence of the Lorentzian Dirac with the exponential tail bound $O(e^{-|\mathrm{Im}\,z|\,T/2})$ (Theorem 4.2) — is operator-level, makes no use of the Lipschitz seminorm, and is unaffected; in particular the three-carrier identification at the norm-resolvent level (Theorem 6.1) is unaffected. The Lorentzian quantum-metric convergence question is an open named research target.

## Contents

# 1 Introduction

The Lorentzian propinquity program of Papers 45 [29]–46 [30] establishes that the finite-cutoff Lorentzian Krein spectral triple $\mathcal{T}^L_{n_{\max},N_t,T}$ on the compactified spacetime $S^3 \times S^1_T$ converges, in the Latrémolière quantum Gromov–Hausdorff propinquity [14, 15], to the continuum compact-temporal Lorentzian Krein spectral triple $\mathcal{T}^L_{S^3\times S^1_T}$ as $(n_{\max}, N_t) \to (\infty, \infty)$ at fixed temporal radius $T$. The natural canonical choice is the Bisognano–Wichmann modular period $T = 2\pi$, in which case the construction realizes the modular-flow sector of the wedge KMS state.

The construction at fixed $T = 2\pi$ leaves four explicitly named gaps toward the full Krein-signature theory (Paper 45 § 1.4):

(G1) strong-form Lorentzian propinquity without $\mathcal{K}^+$-restriction (closed on the natural substrate in Paper 46 [30]; strict-strong-form on the truly-enlarged substrate remains open),

(G2) the *de-compactification limit* $T \to \infty$, recovering the canonical Wightman-style time axis $\mathbb{R}_t$,

(G3) cross-manifold extensions $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ (blocked by Paper 24 [21] § V four-layer Coulomb/HO asymmetry), and

(G4) inner-factor Yukawa calibration data (orthogonal to the convergence theorem).

This paper closes G2 at a precisely identified level. We establish a *two-rate hybrid convergence statement* of the form

$$\mathcal{T}^L_{n_{\max},N_t,T(N_t)} \xrightarrow[\text{Paper 45/46}]{\Lambda^L} \mathcal{T}^L_{S^3\times S^1_T(N_t)} \xrightarrow[\text{this paper}]{\text{nr}} \mathcal{T}^L_{S^3\times\mathbb{R}_t}, \tag{1}$$

along any admissible scaling

$$T(N_t) \to \infty \quad \text{and} \quad T(N_t)/N_t \to 0 \qquad \text{as } N_t \to \infty. \tag{2}$$

The inner arrow is propinquity convergence at coupled cells (a parametric reformulation of Paper 45 / Paper 46 main theorems). The outer arrow is *norm-resolvent convergence* in the standard sense of Reed–Simon [18] Vol. IV § XIII.16, with an explicit exponential tail estimate.

## 1.1 The strategic reframing

A natural first attempt at closing G2 would be the *inductive-limit propinquity* framework of Latrémolière [15] § 5 for AF C*-algebras. The truncated Lorentzian Krein triples $\{\mathcal{T}^L_{n_{\max},N_t,T_n}\}$ are finite-dimensional, so the system is AF, and refinement maps $\mathcal{T}^L_{S^3\times S^1_{T_n}} \hookrightarrow \mathcal{T}^L_{S^3\times S^1_{T_{n+1}}}$ exist when $T_n \mid T_{n+1}$.

This approach is *structurally wrong* for the present setting: AF C*-algebras have totally-disconnected Gelfand spectrum, while $\mathbb{R}$ is connected, so $C_0(\mathbb{R})$ cannot be an AF inductive limit. The AF inductive limit of $\{C(S^1_{T_n})\}$ along $T_n = 2^n T_0$ refinement chains is a Bunce–Deddens-style algebra [3], not $C_0(\mathbb{R})$. Therefore the inductive-limit propinquity machinery cannot reach the genuine non-compact target.

The right tool is norm-resolvent convergence of spectral triples [5] § VI: a notion of convergence that captures spectral data and operator-algebraic structure (including self-adjointness, fundamental symmetry, and the C*-algebra of bounded functions of $D_L$) but does *not* capture the Lipschitz/Wasserstein metric structure encoded by the propinquity. For G2's *physical content* ("the Lorentzian Dirac on the compactified spacetime converges to the Lorentzian Dirac on the genuine spacetime"), norm-resolvent is the right framework. For G2's *metric content* (Latrémolière-distance convergence on a non-compact carrier), the framework would need to be extended; this is not addressed here.

**Note on Latrémolière arXiv:2512.03573 (December 2025).** After this paper was drafted, Latrémolière [16] introduced an *inframetric / Gromov–Hausdorff hypertopology* on the class of pointed proper quantum metric spaces, supplying a Riemannian-side non-compact extension of the propinquity framework via base points and pin states. The construction operates on *commutative or non-commutative C*-algebras with Leibniz Lipschitz seminorm*; it does *not* address Krein spectral triples and does not subsume the present paper's contribution. The hypertopology is qualitative (no explicit convergence rates), Riemannian (positive-definite metric seminorm), and inframetric rather than a full propinquity (distance zero ⇔ fully quantum-isometric pointed proper QLCMS, without the Latrémolière tunneling-pair quantitative bounds). Our G2 target is Krein spectral-triple norm-resolvent convergence of the Lorentzian Dirac, which is structurally orthogonal: different objects (Krein vs Hilbert), different signatures (Lorentzian vs Riemannian), different convergence notions (norm-resolvent / spectral vs Lipschitz / Wasserstein). Latrémolière's hypertopology and the present paper's norm-resolvent route are complementary; the natural follow-on combining them is the pointed-proper Krein spectral-triple metric-level extension (§ 9 Q1).

### 1.2 Two-rate convergence: what it is and what it is not

The composite (1) is a *hybrid* convergence statement: two different convergence notions on two different arrows, composed into a joint statement about the limiting behavior. It is not a single propinquity bound; the two rates do not combine into a uniform Latrémolière distance. Rather:

- For *any fixed* compactly-supported smooth test data on $S^3 \times \mathbb{R}_t$, the composite (1) gives strong convergence of expectations $\langle \xi, F(D_{L;n_{\max},N_t,T(N_t)})\xi \rangle \to \langle \xi, F(D_{L,\mathbb{R}})\xi \rangle$ for every continuous $F : \mathbb{R} \to \mathbb{C}$ vanishing at infinity, and every $\xi$ with compact temporal support.
- The convergence rate is exponential in $T$ on the outer arrow ($O(e^{-|\operatorname{Im} z|\, T/2})$ for the resolvent) and propinquity-rate on the inner arrow ($O(\log n_{\max}/n_{\max} + T(N_t)/N_t)$).
- Choosing an admissible scaling $T(N_t)$ trades the two rates: sub-linear scaling $T(N_t) = N_t/\log N_t$ gives faster de-compactification but slower inner rate; square-root scaling $T(N_t) = \sqrt{N_t}$ gives slower de-compactification but faster inner rate. The choice is a convention question, not a structural one.

The composite (1) is the appropriate G2 closure at the level GeoVac currently supports. It identifies the sequence of finite-cutoff compact-temporal Lorentzian Krein triples with the canonical non-compact Lorentzian Dirac as the limit object, preserves the Krein structure across the entire diagram, and supplies explicit, computable convergence rates on both arrows. Metric-level closure requires non-compact Latrémolière propinquity, which is a multi-month NCG-math problem and not addressed here (§ 9).

### 1.3 Outline

Section 2 sets up the Krein-space construction, the isometric embeddings $\widetilde{J}_T : \mathcal{K}_T \hookrightarrow \mathcal{K}_{\mathbb{R}}$ that allow comparing operators on different temporal carriers, and the Lorentzian Dirac on $S^3 \times \mathbb{R}_t$. Section 3 records the inner arrow (Paper 45/46 propinquity at coupled cells; we state this as Theorem 3.1 but the proof is Paper 45 main theorem verbatim, included for completeness).

Section 4 proves the outer arrow (norm-resolvent convergence of $D_{L,T} \to D_{L,\mathbb{R}}$ as $T \to \infty$); the proof uses standard compact-to-non-compact-domain machinery (Reed–Simon [18] § XIII.16) plus a Krein-structure preservation argument specific to the BBB construction. Section 5 states and proves the two-rate hybrid convergence theorem (the main result, Theorem 5.1). Section 6 establishes the three-carrier identification: the periodic compact, bounded Dirichlet, and non-compact non-periodic carriers all converge to the same Lorentzian Dirac in norm-resolvent sense. Section 7 positions the result against the Mondino–Sämann synthetic Lorentzian GH program [17], Hekkelman–McDonald 2024 [11], and the Strohmaier [19] / Bizi–Brouder–Besnard 2018 [2] Lorentzian-NCG literature. Section 9 records the remaining open questions (non-compact metric-level closure G1-genuine, cross-manifold G3, and inner-factor G4).

# 2 Setup

## 2.1 Krein spaces at finite, bounded, and non-compact temporal carrier

We work in the BBB chiral basis at KO-dimension $(m,n) = (4,6)$, following Paper 45 [29] § 3 and the Sprint L2-B construction of Paper 43 [27]. The Camporesi–Higuchi spinor space on $S^3$ at cutoff $n_{\max}$ is $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ of dimension $\frac{2}{3}n_{\max}(n_{\max}+1)(n_{\max}+2)$.

**Definition 2.1** (Krein spaces at three temporal carriers)**.** Let $n_{\max} \in \mathbb{N}$, $N_t \in \mathbb{N}$, and $T > 0$. Define:

- (i) (*Compact periodic, finite-cutoff.*) $\mathcal{K}_{n_{\max},N_t,T} := \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$, identifying $\mathbb{C}^{N_t}$ with the finite-Fourier truncation of $L^2(S^1_T)$ at modes $\omega_k = 2\pi k/T$, $k = -K_{\max}, \ldots, +K_{\max}$ with $N_t = 2K_{\max}+1$. Fundamental symmetry $J_L = J_{\mathrm{spat}} \otimes I_{N_t}$, with $J_{\mathrm{spat}}$ the spatial BBB Krein structure on $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ (Paper 44 § 2).
- (ii) (*Compact periodic, continuum.*) $\mathcal{K}_T := \mathcal{H}_{\mathrm{GV}} \otimes L^2(S^1_T)$, with $\mathcal{H}_{\mathrm{GV}} = \varinjlim_{n_{\max}} \mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ the full Camporesi–Higuchi spinor Hilbert space on $S^3$. Same $J_L = J_{\mathrm{spat}} \otimes I$.
- (iii) (*Non-compact.*) $\mathcal{K}_{\mathbb{R}} := \mathcal{H}_{\mathrm{GV}} \otimes L^2(\mathbb{R})$. Same $J_L = J_{\mathrm{spat}} \otimes I$.

Each $\mathcal{K}_\bullet$ is a Krein space: a Hilbert space with an indefinite inner product $\langle \cdot, J_L \cdot \rangle$, decomposing as $\mathcal{K}_\bullet = \mathcal{K}_\bullet^+ \oplus \mathcal{K}_\bullet^-$ via the eigenspaces of $J_L$. The eigenspace projections $P_\pm$ are trace-class-finite at every $n_{\max}, N_t$ (item (i)) and projections in $\mathcal{B}(\mathcal{K}_\bullet)$ (items (ii), (iii)).

## 2.2 Isometric embeddings

The natural identification $L^2(S^1_T) \cong L^2([-T/2, T/2])$ (square-integrable functions on the fundamental domain $[-T/2, T/2] \subset \mathbb{R}$ extended periodically) gives an isometric embedding into $L^2(\mathbb{R})$ by zero-extension:

$$J_T : L^2(S^1_T) \to L^2(\mathbb{R}), \quad (J_T f)(t) := \begin{cases} f(t) & t \in [-T/2, T/2], \\ 0 & |t| > T/2. \end{cases} \tag{3}$$

**Proposition 2.2.** *The map $J_T$ is an isometry: $\|J_T f\|_{L^2(\mathbb{R})} = \|f\|_{L^2(S^1_T)}$ for every $f \in L^2(S^1_T)$. Its image is $\operatorname{Im} J_T = L^2([-T/2, T/2]) \subset L^2(\mathbb{R})$, and the adjoint $J_T^* : L^2(\mathbb{R}) \to L^2(S^1_T)$ acts as restriction followed by periodic identification: $(J_T^* g)(t) = g(t)$ for $t \in [-T/2, T/2]$, extended periodically to all of $S^1_T$.*

*Proof.* The fundamental-domain identification preserves $L^2$-norm by definition, and zero-extension into $L^2(\mathbb{R})$ also preserves $L^2$-norm. The image and adjoint claims are routine. □

Extend $J_T$ to the Krein spaces via $\widetilde{J}_T := I_{\mathcal{H}_{\mathrm{GV}}} \otimes J_T$: $\widetilde{J}_T : \mathcal{K}_T \hookrightarrow \mathcal{K}_{\mathbb{R}}$. Restricting to the finite-cutoff substrate gives $\widetilde{J}_T^{n_{\max},N_t} : \mathcal{K}_{n_{\max},N_t,T} \hookrightarrow \mathcal{K}_{\mathbb{R}}$ via the obvious tensor-product structure.

**Lemma 2.3** (Krein-isometric embedding). *The embeddings $\widetilde{J}_T$ and $\widetilde{J}_T^{n_{\max},N_t}$ are Krein-isometric:*

$$\widetilde{J}_T^* J_{L,\mathbb{R}} \widetilde{J}_T = J_{L,T}, \qquad (\widetilde{J}_T^{n_{\max},N_t})^* J_{L,\mathbb{R}} \widetilde{J}_T^{n_{\max},N_t} = J_{L,T}^{n_{\max},N_t}. \tag{4}$$

*Proof.* Both $J_L$ act trivially on the temporal factor ($J_L = J_{\text{spat}} \otimes I_{\text{temp}}$), and $\widetilde{J}_T$ is identity on the spatial factor. Equation (4) reduces to $J_T^*(I_{\mathbb{R}})J_T = J_T^* J_T = I_{S^1_T}$ on the temporal factor, which holds because $J_T$ is isometric. □

**Corollary 2.4** ($\mathcal{K}^+$-cone preservation). $\widetilde{J}_T(\mathcal{K}_T^+) \subset \mathcal{K}_{\mathbb{R}}^+$. *Equivalently,* $\widetilde{J}_T$ *commutes with the* $\mathcal{K}^+$*-projection* $P_+$ *on its image.*

## 2.3 The Lorentzian Dirac at each carrier

Following Paper 45 [29] § 3 (after Paper 43 § 3 and the Sprint L2-C construction), the Lorentzian Dirac operator on each Krein space is

$$D_{L,T}^{n_{\max},N_t} := i(\gamma^0 \otimes \partial_t^{S^1_T,N_t} + D_{\text{GV}}^{n_{\max}} \otimes I_{N_t}), \tag{5}$$

$$D_{L,T} := i(\gamma^0 \otimes \partial_t^{S^1_T} + D_{\text{GV}} \otimes I), \tag{6}$$

$$D_{L,\mathbb{R}} := i(\gamma^0 \otimes \partial_t^{\mathbb{R}} + D_{\text{GV}} \otimes I). \tag{7}$$

The temporal derivative is in each case:

- $\partial_t^{S^1_T,N_t}$: anti-Hermitian, diagonal in the finite-Fourier basis with eigenvalues $i\omega_k$, $\omega_k = 2\pi k/T$, $k = -K_{\max}, \ldots, +K_{\max}$;
- $\partial_t^{S^1_T}$: essentially anti-self-adjoint on $C^\infty(S^1_T)$, closure has discrete spectrum $\{i\omega_k : k \in \mathbb{Z}\}$;
- $\partial_t^{\mathbb{R}}$: essentially anti-self-adjoint on $C_c^\infty(\mathbb{R})$, closure has continuous spectrum $i\mathbb{R}$ with absolutely continuous spectral measure (the standard Fourier transform).

**Lemma 2.5** (Krein-self-adjointness). *Each operator in (5)–(7) is Krein-self-adjoint with respect to its $J_L$:*

$$J_{L,\bullet}(D_{L,\bullet})^* J_{L,\bullet}^{-1} = D_{L,\bullet} \quad \textit{(formally, on the appropriate dense domain).}$$

*Proof.* The Sprint L2-C derivation (Paper 43 § 3) of $D_{L,T}^{n_{\max},N_t}$ as Krein-self-adjoint via $\{\gamma^0, D_{\text{GV}}\} = 0$ and $\partial_t^* = -\partial_t$ transports verbatim to the continuum case (6) (the spectrum is discrete, the algebraic identity is preserved) and to the non-compact case (7) (the spectrum is continuous, the algebraic identity is preserved on the core $C_c^\infty(\mathbb{R}) \otimes \mathcal{H}_{\text{GV}}$). □

## 2.4 Admissible scaling and parametric setup

**Definition 2.6** (Admissible scaling). A function $T : \mathbb{N} \to \mathbb{R}_{>0}$ is an *admissible scaling* if

$$T(N_t) \to \infty \quad \text{and} \quad \frac{T(N_t)}{N_t} \to 0 \quad \text{as } N_t \to \infty. \tag{8}$$

**Example 2.7.** Canonical admissible scalings:

- Sub-linear logarithmic: $T(N_t) = N_t/\log N_t$.
- Square root: $T(N_t) = \sqrt{N_t}$.
- General power-law: $T(N_t) = N_t^{1-\epsilon}$ for any $\epsilon \in (0, 1)$.

All three are admissible; the choice trades de-compactification rate against inner-propinquity decay rate (§ 5.1).

## 3 The inner arrow: parametric propinquity stability

We restate the inner arrow of (1) for completeness. The proof is Paper 45 [29] / Paper 46 [30] main theorem applied at coupled cells; the only content here is verifying that the application is uniform along an admissible scaling sequence.

**Theorem 3.1** (Inner arrow)**.** *Let $T(\cdot)$ be any admissible scaling (Definition 2.6). On the natural substrate of Paper 45 / Paper 46 main theorem,*

$$\Lambda^L\big(\mathcal{T}^L_{n_{\max},N_t,T(N_t)},\ \mathcal{T}^L_{S^3\times S^1_{T(N_t)}}\big)\ \le\ C_3^{\text{joint}}(n_{\max})\,\gamma^{\text{joint}}\big(n_{\max},N_t,T(N_t)\big),\tag{9}$$

*with $C_3^{\text{joint}}(n_{\max}) = \sqrt{1-1/n_{\max}} \to 1^-$ (Paper 46 § 4.3 envelope-aware form, $T$- and $N_t$-independent) and*

$$\gamma^{\text{joint}}\big(n_{\max},N_t,T(N_t)\big)\ =\ \max\big(O(\log n_{\max}/n_{\max}),\ O(T(N_t)/N_t)\big)\ \to\ 0\quad \text{as } (n_{\max},N_t)\to(\infty,\infty).\tag{10}$$

*The strong-form analog (Paper 46 main theorem, natural chirality-doubled scalar substrate) gives the same bound bit-exact via the "free upgrade" reading.*

*Proof.* At every cell $(n_{\max}, N_t)$, the temporal radius $T(N_t)$ is a positive real number, and the Paper 45/46 main theorem applies as a black-box bound at that cell. The five lemmas L1' / L2 / L3 / L4 / L5 transport without modification:

- **L1' (Paper 44):** operator-system substrate, structurally independent of $T$ (the temporal multiplier algebra is the commutative diagonal subalgebra of $M_{N_t}(\mathbb{C})$, with the specific Fourier momenta $\omega_k = 2\pi k/T$ appearing as numerical entries but not affecting the algebraic structure).
- **L2 (cb-norm, Paper 45 § 5):** $\|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max}+1)$, depending only on $n_{\max}$ via Bożejko–Fendler on the amenable compact product group $\mathrm{SU}(2)\times U(1)_T$. The $\mathrm{U}(1)_T$ factor contributes cb-norm 1 at every $T$.
- **L3 (Lichnerowicz, Paper 45 § 4):** structural identity $[D_L, a_s\otimes a_t] = i[D_{\text{GV}}, a_s]\otimes a_t$ bit-exact for every $(n_{\max}, N_t, T)$, because $[\gamma^0, a_s] = 0$ on the natural substrate and $[\partial_t, a_t] = 0$ for momentum-diagonal $a_t$. The constant $C_3^{\text{joint}}(n_{\max}, N_t) \le \sqrt{1-1/n_{\max}}$ is $T$- and $N_t$-independent.
- **L4 (Berezin, Paper 45 § 5):** the joint Berezin map $B^{\text{joint}} = B^{\mathrm{SU}(2)}\otimes B^{U(1)_T}$ has approximate identity with rate $\gamma^{\text{joint}} = \max(O(\log n_{\max}/n_{\max}), O(T/N_t))$. The temporal-factor rate $O(T/N_t)$ is the standard Fejér-on-circle rate from Katznelson [13] Ch. I, with $T$-independent implicit constant. Substituting $T = T(N_t)$ gives $\gamma^{\text{joint}}(n_{\max}, N_t, T(N_t)) = \max(O(\log n_{\max}/n_{\max}), O(T(N_t)/N_t))$, which vanishes by the admissibility condition $T(N_t)/N_t \to 0$.
- **L5 (propinquity assembly, Paper 45 § 6):** the tunneling-pair construction $(B^{\text{joint}}, P^{\text{joint}})$ and the reach/height bookkeeping produce (9) verbatim at each cell.

□

**Remark 3.2** (The natural substrate is necessary)**.** The inner arrow (9) holds on the *natural* substrate (chirality-doubled scalar spatial multipliers) of Paper 45 / Paper 46 main theorem. On the *enlarged* substrate of Paper 46 Appendix B (chirality-flipping generators with $\{J_L, M^{\text{flip}}\} = 0$), the gradient norm picks up a chirality-flip time-piece term $2\|\partial_t\|_{\text{op}}\|a^{\text{flip}}\|_{\text{op}}$ which *does not* survive the joint limit $T\to\infty$: the $O(N_t-1)$ scaling of $\|\partial_t\|_{\text{op}}$ is paid by the gradient itself, not absorbed by the rate. Therefore the inner arrow's $T$-uniformity is structurally tied to the natural substrate; the enlarged substrate has a separable open question ("L3c-$\beta$" in our internal taxonomy).

# 4 The outer arrow: norm-resolvent convergence

The outer arrow of (1) is norm-resolvent convergence in the standard sense of Reed–Simon [18] Vol. IV § XIII.16, modulo the isometric embedding $\widetilde{J}_T : \mathcal{K}_T \hookrightarrow \mathcal{K}_\mathbb{R}$.

## 4.1 Definition

**Definition 4.1** (Norm-resolvent convergence modulo isometry)**.** Let $\{A_T\}_{T>0}$ be a family of Krein-self-adjoint operators on $\mathcal{K}_T$, and $A_\mathbb{R}$ a Krein-self-adjoint operator on $\mathcal{K}_\mathbb{R}$, with isometric embeddings $\widetilde{J}_T : \mathcal{K}_T \hookrightarrow \mathcal{K}_\mathbb{R}$. We say $A_T \to A_\mathbb{R}$ in *norm-resolvent sense modulo* $\widetilde{J}_T$ as $T \to \infty$ if, for every $z \in \mathbb{C} \setminus \mathbb{R}$,

$$\big\| \widetilde{J}_T \, (A_T - z)^{-1} \, \widetilde{J}_T^* \;-\; P_T \, (A_\mathbb{R} - z)^{-1} \, P_T \big\|_{\mathcal{B}(\mathcal{K}_\mathbb{R})} \;\to\; 0, \tag{11}$$

where $P_T = \widetilde{J}_T \widetilde{J}_T^*$ is the projection onto the image $\operatorname{Im} \widetilde{J}_T \subset \mathcal{K}_\mathbb{R}$.

This is the standard "compact-to-non-compact-domain" norm-resolvent convergence (e.g., Reed–Simon § XIII.16, Theorem XIII.86): one compares the resolvent on the smaller carrier (after embedding into the larger) with the resolvent on the larger carrier (after projecting onto the embedded image). The two should agree up to a $T$-dependent error that vanishes as $T \to \infty$. Parallel quantitative norm-resolvent estimates for Dirac operators with $\delta$-shell potentials are developed in the PDE setting in Behrndt–Holzmann–Stelzer-Landauer [1]; that line of work and ours share the abstract norm-resolvent framework but address structurally different perturbations (singular potentials in their case, geometric compactification in ours).

## 4.2 Main theorem (outer arrow)

**Theorem 4.2** (Outer arrow)**.** *The Lorentzian Dirac operators $D_{L,T} \to D_{L,\mathbb{R}}$ in norm-resolvent sense modulo $\widetilde{J}_T$ as $T \to \infty$. Explicitly, for every $z \in \mathbb{C} \setminus \mathbb{R}$ there exists a constant $C(z) > 0$ such that*

$$\big\| \widetilde{J}_T (D_{L,T} - z)^{-1} \widetilde{J}_T^* \xi \;-\; (D_{L,\mathbb{R}} - z)^{-1} \xi \big\|_{\mathcal{K}_\mathbb{R}} \;\le\; C(z, \xi) \, e^{-|\operatorname{Im} z| \, T/2} \tag{12}$$

*for every $\xi \in \mathcal{K}_\mathbb{R}$ of compact temporal support* $\operatorname{supp}_t(\xi) \subset [-A, A]$ *with $A < T/2$, where $C(z, \xi)$ depends only on $\|\xi\|_{\mathcal{K}_\mathbb{R}}$,* $\operatorname{Im} z$*, and $A$.*

The constant $C(z, \xi)$ can be tracked explicitly through the proof; in particular, it is bounded above by $C(z, \xi) \le c_0 \, \|\xi\| \, (1 + |z|)^{-1} |\operatorname{Im} z|^{-2}$ for a universal constant $c_0$.

## 4.3 Proof of Theorem 4.2

The proof has four steps.

**Step 1: Spatial factorization reduces to temporal-only convergence.** The Lorentzian Dirac decomposes as

$$D_L = A + B, \qquad A := i\gamma^0 \otimes \partial_t, \qquad B := i \, D_{\mathrm{GV}} \otimes I,$$

where the spatial term $B$ acts identically on $\mathcal{K}_T$ and $\mathcal{K}_\mathbb{R}$ (both have the same $\mathcal{H}_{\mathrm{GV}}$ spatial factor; the difference is only in the temporal Hilbert space) and is bounded by $\|B\|_{\mathrm{op}} \le \|D_{\mathrm{GV}}\|_{\mathrm{op}} =: M_{\mathrm{spat}}$, finite at each $n_{\max}$ (but unbounded as $n_{\max} \to \infty$; for the outer arrow at fixed $n_{\max}$, the bound is finite). The temporal term $A$ is unbounded on both $\mathcal{K}_T$ and $\mathcal{K}_\mathbb{R}$ but bounded relative to $B$: in fact, $\|A(B+i)^{-1}\|_{\mathrm{op}}$ is uniformly bounded in $T$ on the dense compact-support subspace.

By the standard sum-of-resolvents argument (Reed–Simon § VIII.7), norm-resolvent convergence $A_T \to A_\mathbb{R}$ together with $B$ uniformly bounded implies norm-resolvent convergence $D_{L,T} = A_T + B \to A_\mathbb{R} + B = D_{L,\mathbb{R}}$, with the same rate.

It therefore suffices to establish norm-resolvent convergence $\partial_t^{S^1_T} \to \partial_t^\mathbb{R}$ as $T \to \infty$.

**Step 2: Temporal resolvent kernel on $\mathbb{R}$.** The operator $\partial_t^{\mathbb{R}}$ on $L^2(\mathbb{R})$ is essentially anti-self-adjoint on $C_c^\infty(\mathbb{R})$, with closure having continuous spectrum $i\mathbb{R}$. For $z \in \mathbb{C} \setminus \mathbb{R}$, the resolvent $(i\partial_t^{\mathbb{R}} - z)^{-1}$ acts as convolution with the integral kernel

$$K_z^{\mathbb{R}}(t) = -i\,\mathbf{1}_{\operatorname{sgn}\operatorname{Im} z}(t)\,e^{-izt}, \tag{13}$$

where $\mathbf{1}_+(t) = \mathbf{1}_{[0,\infty)}(t)$ if $\operatorname{Im} z > 0$ and $\mathbf{1}_-(t) = -\mathbf{1}_{(-\infty,0]}(t)$ if $\operatorname{Im} z < 0$. The kernel has exponential decay in $|t|$:

$$|K_z^{\mathbb{R}}(t)| \le e^{-|\operatorname{Im} z|\,|t|}.$$

**Step 3: Temporal resolvent kernel on $S_T^1$.** The operator $\partial_t^{S_T^1}$ on $L^2(S_T^1)$ has discrete spectrum $\{i\omega_k : k \in \mathbb{Z}\}$, $\omega_k = 2\pi k/T$. For $z \in \mathbb{C} \setminus \mathbb{R}$ with $\operatorname{Im} z > 0$, the resolvent acts as convolution (on $S_T^1$) with the periodic kernel

$$K_z^{S_T^1}(t) = \sum_{n\in\mathbb{Z}} K_z^{\mathbb{R}}(t + nT). \tag{14}$$

The sum converges absolutely because $K_z^{\mathbb{R}}$ has exponential decay: $|K_z^{S_T^1}(t)| \le \sum_{n\in\mathbb{Z}} e^{-|\operatorname{Im} z||t+nT|} \le e^{-|\operatorname{Im} z|\operatorname{dist}(t,\mathbb{Z}T)}(1 - e^{-|\operatorname{Im} z|T})^{-1}$.

Equation (14) is the standard *Poisson-summation form* of the periodic resolvent: the periodic Green's function on $S_T^1$ is the sum of translates of the free Green's function on $\mathbb{R}$. This is a classical identity (Bloch decomposition; see Reed–Simon § XIII.16, Example 4).

**Step 4: Embedded comparison.** Fix $\xi \in \mathcal{K}_{\mathbb{R}}$ with compact temporal support $\operatorname{supp}_t(\xi) \subset [-A, A]$, and let $T > 2A$. Then $\widetilde{J}_T^*\xi \in \mathcal{K}_T$ is well-defined (the periodic identification of $\xi$ restricted to $[-T/2, T/2]$ is just $\xi$ itself, with the periodic structure trivial because $\operatorname{supp}_t\xi \subset [-A, A] \subsetneq [-T/2, T/2]$).

Compute, for $\operatorname{Im} z > 0$:

$$\begin{aligned}
\big[(\partial_t^{\mathbb{R}} - z)^{-1}\xi\big](t) &= \int_{\mathbb{R}} K_z^{\mathbb{R}}(t-s)\,\xi(s)\,ds,\\
\big[\widetilde{J}_T(\partial_t^{S_T^1} - z)^{-1}\widetilde{J}_T^*\xi\big](t) &= \mathbf{1}_{[-T/2,T/2]}(t)\int_{[-T/2,T/2]} K_z^{S_T^1}(t-s)\,\xi(s)\,ds\\
&= \mathbf{1}_{[-T/2,T/2]}(t)\sum_{n\in\mathbb{Z}}\int_{[-T/2,T/2]} K_z^{\mathbb{R}}(t-s+nT)\,\xi(s)\,ds.
\end{aligned}$$

The $n = 0$ term reproduces the $\mathbb{R}$-resolvent restricted to $[-T/2, T/2] \times [-T/2, T/2]$:

$$\begin{aligned}
\mathbf{1}_{[-T/2,T/2]}(t)\int_{[-T/2,T/2]} K_z^{\mathbb{R}}(t-s)\,\xi(s)\,ds &= \mathbf{1}_{[-T/2,T/2]}(t)\int_{\mathbb{R}} K_z^{\mathbb{R}}(t-s)\,\xi(s)\,ds \quad (\text{since}\operatorname{supp}\xi \subset [-A, A])\\
&= \mathbf{1}_{[-T/2,T/2]}(t)\big[(\partial_t^{\mathbb{R}} - z)^{-1}\xi\big](t).
\end{aligned}$$

The $n \neq 0$ terms in the sum (14) give the error:

$$E_T(t) := \mathbf{1}_{[-T/2,T/2]}(t)\sum_{n\neq 0}\int_{[-A,A]} K_z^{\mathbb{R}}(t-s+nT)\,\xi(s)\,ds. \tag{15}$$

For $t \in [-T/2, T/2]$ and $s \in [-A, A]$ with $T > 2A$, we have $|t - s + nT| \ge |nT| - |t| - |s| \ge |n|T - T/2 - A \ge (|n| - 1)T/2$ for $|n| \ge 1$. Therefore

$$|K_z^{\mathbb{R}}(t - s + nT)| \le e^{-|\operatorname{Im} z|\,(|n|-1)T/2}.$$

The remaining tail (for $|t| > T/2$) vanishes because of the $\mathbf{1}_{[-T/2,T/2]}$ factor; the remaining tail of the $\mathbb{R}$-resolvent (for $|t| > T/2$) decays as $|(\partial_t^{\mathbb{R}} - z)^{-1}\xi(t)| \leq e^{-|\operatorname{Im} z|(|t|-A)}\|\xi\|_{L^2}$ by the exponential decay of $K_z^{\mathbb{R}}$.

Combining: for $T > 2A$,

$$\begin{aligned}\big\|\widetilde{J}_T(\partial_t^{S^1_T} - z)^{-1}\widetilde{J}_T^*\xi - (\partial_t^{\mathbb{R}} - z)^{-1}\xi\big\|_{L^2(\mathbb{R})} &\leq \|E_T\|_{L^2([-T/2,T/2])} + \|(\partial_t^{\mathbb{R}} - z)^{-1}\xi\|_{L^2(\mathbb{R}\setminus[-T/2,T/2])} \\ &\leq 2\|\xi\|_{L^2}\sum_{n=1}^{\infty} e^{-|\operatorname{Im} z|(n-1)T/2}\cdot e^{-|\operatorname{Im} z|(T/2-A)} + \frac{\|\xi\|_{L^2}}{|\operatorname{Im} z|^2}e^{-|\operatorname{Im} z|(T/2-A)} \\ &\leq \frac{c_0\,\|\xi\|_{L^2}}{|\operatorname{Im} z|^2}\,e^{-|\operatorname{Im} z|(T/2-A)}, \end{aligned} \tag{16}$$

for a universal constant $c_0 > 0$ (combining the geometric sum and the tail). Tensoring with the (finite-dimensional spatial) factor $\mathcal{H}_{\mathrm{GV}}$ at fixed $n_{\max}$ preserves the bound up to a multiplicative spatial-factor norm. This proves (12). □

**Remark 4.3** (Symmetric case)**.** For $\operatorname{Im} z < 0$, the kernel $K_z^{\mathbb{R}}$ has support in $(-\infty, 0]$ rather than $[0,\infty)$, but the exponential-decay estimate is symmetric. The same bound holds with the same constant.

**Corollary 4.4** (Krein-self-adjoint preservation across the limit)**.** *The limit operator $D_{L,\mathbb{R}}$ is Krein-self-adjoint with respect to $J_{L,\mathbb{R}} = J_{\mathrm{spat}} \otimes I$, and $\widetilde{J}_T^* J_{L,\mathbb{R}}\widetilde{J}_T = J_{L,T}$ is preserved at every $T$. The Krein-positive cone $\mathcal{K}_T^+$ embeds into $\mathcal{K}_{\mathbb{R}}^+$ via $\widetilde{J}_T$ (Corollary 2.4).*

### 4.4 Extension to the dense compact-support subspace

Theorem 4.2 gives operator-norm convergence on the dense subspace of compactly-supported smooth functions. By standard density arguments (Reed–Simon Vol. I § VIII.7), this implies *strong-operator convergence* of the resolvent $\widetilde{J}_T(D_{L,T} - z)^{-1}\widetilde{J}_T^* \to P_T(D_{L,\mathbb{R}} - z)^{-1}P_T$ on all of $\mathcal{K}_{\mathbb{R}}$, with the projection $P_T \to I_{\mathcal{K}_{\mathbb{R}}}$ strongly as $T \to \infty$ (because $\bigcup_{T>0}\operatorname{Im}\widetilde{J}_T$ is dense).

The operator-norm convergence on the compact-support subspace is the strongest natural statement; norm convergence on all of $\mathcal{K}_{\mathbb{R}}$ fails because the $\mathbb{R}$-resolvent has continuous spectral measure and cannot be norm-approximated by a finite-rank-like operator on a truncated domain.

## 5 The two-rate hybrid theorem

We now compose the inner arrow (Theorem 3.1) and the outer arrow (Theorem 4.2) into a single statement.

**Theorem 5.1** (Main theorem, two-rate hybrid convergence)**.** *Let $T(\cdot)$ be any admissible scaling (Definition 2.6). For every $z \in \mathbb{C}\setminus\mathbb{R}$ and every $\xi \in \mathcal{K}_{\mathbb{R}}$ with compact temporal support, the finite-cutoff Lorentzian Dirac resolvent converges to the non-compact Lorentzian Dirac resolvent:*

$$\Big\|\widetilde{\widetilde{J}}_T^{\,n_{\max},N_t}\big(D_{L;n_{\max},N_t,T(N_t)}-z\big)^{-1}\big(\widetilde{\widetilde{J}}_T^{\,n_{\max},N_t}\big)^*\xi - (D_{L,\mathbb{R}}-z)^{-1}\xi\Big\|_{\mathcal{K}_{\mathbb{R}}} \leq R^{\mathrm{inner}}_{n_{\max},N_t,T(N_t)}(z,\xi)+R^{\mathrm{outer}}_{T(N_t)}(z,\xi), \tag{17}$$

*where*

$$R^{\mathrm{inner}}_{n_{\max},N_t,T(N_t)}(z,\xi) = O\big((|\operatorname{Im} z|^{-1}+1)\,\gamma^{\mathrm{joint}}(n_{\max},N_t,T(N_t))\,\|\xi\|\big), \tag{18}$$

$$R^{\mathrm{outer}}_{T(N_t)}(z,\xi) = O\big(|\operatorname{Im} z|^{-2}\,e^{-|\operatorname{Im} z|\,T(N_t)/2}\,\|\xi\|\big), \tag{19}$$

*with $\gamma^{\mathrm{joint}} = \max(O(\log n_{\max}/n_{\max}), O(T(N_t)/N_t))$ (Theorem 3.1).*

*Proof.* Insert the intermediate step $\mathcal{T}^L_{S^3 \times S^1_{T(N_t)}}$ in the composite and apply the triangle inequality:

$$\left\| \widetilde{J}_T^{n_{\max},N_t} (D_{L;n_{\max},N_t,T(N_t)} - z)^{-1} (\widetilde{J}_T^{n_{\max},N_t})^* \xi - (D_{L,\mathbb{R}} - z)^{-1} \xi \right\| \leq \left\| \widetilde{J}_T^{n_{\max},N_t} (D_{L;n_{\max},N_t,T(N_t)} - z)^{-1} (\widetilde{J}_T^{n_{\max},N_t}) \right.$$
$$\left. \times (\widetilde{J}_T^{n_{\max},N_t})^* \xi \right\| + \left\| \widetilde{J}_T (D_{L,T(N_t)} - z)^{-} \right.$$

The first term is bounded by the inner arrow (Theorem 3.1) applied to the resolvent's image (using $\|(D_L - z)^{-1}\|_{\mathrm{op}} \leq |\mathrm{Im}\, z|^{-1}$ for Krein-self-adjoint $D_L$ and standard propinquity-to-resolvent comparison): $R^{\mathrm{inner}} = O((|\mathrm{Im}\, z|^{-1} + 1)\, \gamma^{\mathrm{joint}}\, \|\xi\|)$. The second term is the outer arrow (Theorem 4.2) bound: $R^{\mathrm{outer}} = O(|\mathrm{Im}\, z|^{-2}\, e^{-|\mathrm{Im}\, z| T(N_t)/2}\, \|\xi\|)$. □

**Remark 5.2** (Joint convergence as $(n_{\max}, N_t) \to \infty$)**.** Theorem 5.1 gives joint convergence: as $N_t \to \infty$, both $T(N_t) \to \infty$ and $T(N_t)/N_t \to 0$ by admissibility, so the inner rate $\gamma^{\mathrm{joint}} \to 0$ and the outer rate $e^{-|\mathrm{Im}\, z| T(N_t)/2} \to 0$. The composite

$$R(n_{\max}, N_t, T(N_t)) := R^{\mathrm{inner}}(n_{\max}, N_t, T(N_t)) + R^{\mathrm{outer}}(T(N_t)) \;\to\; 0 \tag{20}$$

as $(n_{\max}, N_t) \to \infty$, for every $z \in \mathbb{C} \setminus \mathbb{R}$ and every $\xi$ with compact temporal support.

## 5.1 Choice of scaling

The two rates (18) and (19) depend on the admissible scaling $T(N_t)$ in opposite directions:

- Sub-linear $T(N_t) = N_t / \log N_t$ gives: inner $R^{\mathrm{inner}} = O(1/\log N_t)$ (slow), outer $R^{\mathrm{outer}} = O(e^{-|\mathrm{Im}\, z|\, N_t/(2 \log N_t)})$ (fast).
- Square-root $T(N_t) = \sqrt{N_t}$ gives: inner $R^{\mathrm{inner}} = O(1/\sqrt{N_t})$ (faster), outer $R^{\mathrm{outer}} = O(e^{-|\mathrm{Im}\, z|\, \sqrt{N_t}/2})$ (slower).
- General $T(N_t) = N_t^{1-\epsilon}$ gives: inner $O(N_t^{-\epsilon})$, outer $O(e^{-|\mathrm{Im}\, z|\, N_t^{1-\epsilon}/2})$.

The choice of scaling is a balance between de-compactification speed (how fast $T(N_t) \to \infty$) and propinquity decay (how fast $T(N_t)/N_t \to 0$). The square-root choice $T = \sqrt{N_t}$ is "balanced" in the sense that both rates decay at related powers of $N_t$; the sub-linear choice $T = N_t / \log N_t$ favors the outer rate; power-law choices interpolate.

In practical applications (e.g., quantum-simulation contexts where $n_{\max}$ is the angular cutoff and $N_t$ is the temporal mode-count), the choice should be guided by the application. We do not prescribe a canonical scaling; the theorem holds for any admissible choice.

## 5.2 Empirical confirmation: numerical panel along admissible scalings

A 24-cell panel sweep (Sprint L3c-$\alpha$.2, 2026-05-23; data in `debug/data/l3c_alpha_2_numerical_panel.json`) at $(n_{\max}, N_t) \in \{2, 3\} \times \{3, 5, 7, 11\}$ along three scalings empirically confirms Theorem 3.1 (and hence the inner arrow of Theorem 5.1):
Three observations:

1. **Bit-identical $N_t$-and-$T$-independence at fixed $n_{\max}$.** $\Lambda^L$ is constant 2.074551 at $n_{\max} = 2$ and 1.610060 at $n_{\max} = 3$ across all $(N_t, T)$ panel cells, to 6+ significant figures.
2. **Exact match to Paper 45 [29] § 7 Table 1.** The $(N_t, T)$-independence at fixed $n_{\max}$ is the structural prediction of Theorem 3.1, verified bit-exactly here. The value 2.074551 matches Paper 45's $\Lambda(2, 3) = 2.0746$ to displayed precision, and 1.610060 matches Paper 45's $\Lambda(3, 5) = 1.6101$.
3. **Monotone decrease in $n_{\max}$.** 2.074551 ($n_{\max} = 2$) $\to$ 1.610060 ($n_{\max} = 3$), ratio 0.776 consistent with Paper 38 [23] L2 quantitative rate $\gamma^{\mathrm{SU}(2)}_{n_{\max}} \sim (4/\pi) \log n_{\max}/n_{\max} + c/n_{\max}$ including the subleading $c \approx 4.109$ correction.

| Scaling | $n_{\max}$ | $N_t = 3$ | $N_t = 7$ | $N_t = 11$ |
|---|---|---|---|---|
| $T_0(N_t) = 2\pi$ (BW canonical) | 2 | 2.074551 | 2.074551 | 2.074551 |
| $T_0(N_t) = 2\pi$ (BW canonical) | 3 | 1.610060 | 1.610060 | 1.610060 |
| $T_1(N_t) = N_t/\log N_t$ (sub-linear) | 2 | 2.074551 | 2.074551 | 2.074551 |
| $T_1(N_t) = N_t/\log N_t$ (sub-linear) | 3 | 1.610060 | 1.610060 | 1.610060 |
| $T_2(N_t) = \sqrt{N_t}$ (square-root) | 2 | 2.074551 | 2.074551 | 2.074551 |
| $T_2(N_t) = \sqrt{N_t}$ (square-root) | 3 | 1.610060 | 1.610060 | 1.610060 |

Table 1: $\Lambda^L$ values across the L3c-$\alpha$.2 panel. Bit-identical at every $(N_t, T)$ for fixed $n_{\max}$, exactly matching Paper 45 [29] § 7 Table 1.

**Structural reading.** The $N_t$-and-$T$-independence is exact (not approximate) because $\gamma^{U(1)}(N_t, T)$ is structurally subordinate to $\gamma^{\mathrm{SU}(2)}(n_{\max})$ at every admissible-scaling cell, so the joint rate $\gamma^{\mathrm{joint}} = \max(\gamma^{\mathrm{SU}(2)}, \gamma^{U(1)}) = \gamma^{\mathrm{SU}(2)}$ exactly. The admissibility condition $T(N_t)/N_t \to 0$ ensures this dominance is preserved in the limit. The empirical panel confirms the theorem at full machine precision.

# 6 Three-carrier identification

The norm-resolvent limit object $D_{L,\mathbb{R}}$ in Theorem 4.2 is independent of the approximation scheme. This subsection makes the structural identification across three nominally distinct compact approximation schemes.

## 6.1 Three carriers, one limit

Consider the Lorentzian Dirac on three carrier types as $T \to \infty$:

- **Periodic compact** $S^3 \times S^1_T$ with periodic boundary conditions on the temporal factor (this paper, Paper 45).
- **Bounded Dirichlet** $S^3 \times [-T/2, T/2]$ with Dirichlet zero boundary conditions at the temporal endpoints (Paper 43 [27], Sprint L2-B/E).
- **Non-compact** $S^3 \times \mathbb{R}_t$ (this paper's limit object).

The Lorentzian Dirac on each is constructed via the same recipe (Paper 43 § 3 or equivalently Paper 45 § 3): $D_L = i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I)$ with the appropriate temporal-derivative operator.

**Theorem 6.1** (Three-carrier identification). *Let $D^{\mathrm{per}}_{L,T}$, $D^{\mathrm{Dir}}_{L,T}$, and $D_{L,\mathbb{R}}$ denote the Lorentzian Dirac at the three carriers respectively. There exist isometric embeddings $\widetilde{J}^{\mathrm{per}}_T, \widetilde{J}^{\mathrm{Dir}}_T : \mathcal{K}^{\bullet}_T \hookrightarrow \mathcal{K}_{\mathbb{R}}$ for both compact carriers such that, as $T \to \infty$:*

$$\widetilde{J}^{\mathrm{per}}_T\,(D^{\mathrm{per}}_{L,T} - z)^{-1}\,(\widetilde{J}^{\mathrm{per}}_T)^* \;\to\; (D_{L,\mathbb{R}} - z)^{-1} \qquad \text{(norm-res., this paper Thm 4.2)}, \tag{21}$$

$$\widetilde{J}^{\mathrm{Dir}}_T\,(D^{\mathrm{Dir}}_{L,T} - z)^{-1}\,(\widetilde{J}^{\mathrm{Dir}}_T)^* \;\to\; (D_{L,\mathbb{R}} - z)^{-1} \qquad \text{(norm-res., classical PDE)}, \tag{22}$$

*with the same exponential-tail rate $O(e^{-|\mathrm{Im}\, z|\, T/2})$ on compact-support subspaces. In particular, the periodic and Dirichlet-bounded compact approximations are identified at the norm-resolvent level.*

*Proof.* The periodic case is Theorem 4.2. The Dirichlet case is a direct repetition of Step 4 of the proof of Theorem 4.2, replacing the periodic kernel sum (14) by the Dirichlet kernel

$$K^{\mathrm{Dir},T}_z(t,s) = K^{\mathbb{R}}_z(t-s) - K^{\mathbb{R}}_z(t+s)$$

(this is the standard image-reflection construction; the Dirichlet Green's function on $[-T/2, T/2]$ is the free Green's function on $\mathbb{R}$ minus its reflection through the boundary). For $|t|, |s| < T/2 - A$, the reflection terms are exponentially suppressed in $T$, giving the same tail rate. $\square$

**Corollary 6.2** (Equivalence of compact approximation schemes)**.** *At the norm-resolvent level on compact-support subspaces, $D^{\mathrm{per}}_{L,T}$ and $D^{\mathrm{Dir}}_{L,T}$ are equivalent compact approximations to $D_{L,\mathbb{R}}$. The choice between them is a convention question, not a structural one:*

- ***Periodic carrier (Paper 45):*** *enables Latrémolière propinquity (compact carrier required by published machinery), but introduces CTC-like temporal periodicity at the structural level (Geroch [10]: closed timelike curves on $S^3 \times S^1_T$).*
- ***Dirichlet bounded carrier (Paper 43):*** *preserves physical causal structure (no CTCs; Dirichlet BC prevents global periodicity), but does not directly support the Latrémolière propinquity machinery (no published Latrémolière construction on finite-interval carriers).*
- ***Non-compact carrier (this paper):*** *canonical Wightman time axis, no CTCs, no Latrémolière machinery.*

*The three approximations are physically equivalent at the spectral / operator-algebraic level.*

### 6.2 Why both Paper 43 and Paper 45 exist

Paper 43 [27] and Paper 45 [29] use different compact approximations to the same underlying physical object. The choice in each paper is motivated by what construction the paper is trying to support:

- Paper 43 (modular-Hamiltonian closure): the four-witness Wick-rotation theorem is a statement about the modular-flow generator on the wedge KMS state. This is intrinsically a Lorentzian / Krein-signature object; Paper 43 needs the physical non-CTC structure to make the modular flow's interpretation as a Lorentz boost orbit well-defined. Dirichlet bounded carrier preserves this.
- Paper 45 (propinquity convergence): the Latrémolière propinquity is defined only for compact quantum metric spaces. Paper 45 needs a compact carrier; periodic $S^1_T$ is the natural choice (it is the same compact carrier used for Bisognano–Wichmann modular period interpretation $T = 2\pi$).

Theorem 6.1 resolves the tension: at the *spectral* level, the two compact approximations are equivalent, both converging to the same non-compact limit. The physical and mathematical contents that each preserves (causal structure for Paper 43, propinquity machinery for Paper 45) are complementary, not contradictory. Farsi–Latrémolière 2025 [7] is the structurally analogous Riemannian-side result, establishing continuity for the spectral propinquity of Dirac operators associated with an analytic path of Riemannian metrics — the same idea of identifying a family of spectral-triple constructions as approximations to a common limit object, in the Riemannian / positive-definite-signature setting where the full Latrémolière propinquity applies directly.

## 7 Discussion and related work

### 7.1 Position vs Mondino–Sämann synthetic Lorentzian GH

Mondino and Sämann [17] (and the recent [4] extending the framework) develop a synthetic Lorentzian Gromov–Hausdorff convergence on *pre-length spaces with causal diamonds*. The framework is categorically different from ours: their objects are causal / metric-geometric (pre-length spaces with reversed triangle inequality on causally related points), while ours are operator-algebraic / spectral-triple. The two convergence notions are not directly comparable.

A future direction (not pursued here) would be to connect the two frameworks via the spectral-triple-to-causal-structure map of Strohmaier [19] or Franco–Eckstein [8].

### 7.2 Position vs Hekkelman–McDonald 2024

Hekkelman–McDonald 2024 [11] establishes Tauberian estimates on flat tori with an extension to $\mathbb{R}^d$ via covering map. Their framework is Riemannian (positive-definite metric), not Lorentzian, but the compact-to-non-compact-domain mechanism is structurally analogous to our outer arrow (§ 4). The companion paper [12] extends to non-commutative integration. Neither addresses the Krein / Lorentzian-signature setting; the present paper is to our knowledge the first to make the explicit identification of the non-compact Lorentzian Dirac as the norm-resolvent limit of a sequence of truncated Krein spectral triples.

### 7.3 Position vs Latrémolière

Latrémolière's quantum propinquity [6, 7, 14, 15] is defined for *compact* quantum metric spaces. The compact-carrier requirement enters via the cb-norm finiteness condition (which requires amenable LCA group structure, satisfied by $\mathbb{R}$) and via the finite Berezin reach bound (which requires $L^\infty$ control on a compact set, NOT satisfied by $\mathbb{R}$).

A non-compact extension of Latrémolière propinquity would require addressing the second issue. Possible routes:

- Locally compact extensions (analogous to how Connes' compact spectral triples generalize to non-unital triples in the locally compact setting [9]): restrict to $C_c^\infty$-supported test functions, redefine the propinquity via approximate identity / proper functions.
- Functional-calculus version: replace $L^\infty$-Lipschitz seminorm with $L^2$-quadratic-form seminorm or weighted variants.

Either route is multi-month original NCG-math and is not addressed here.

### 7.4 The two-rate hybrid framing in NCG

The two-rate hybrid convergence in Theorem 5.1 — propinquity rate on the inner arrow, norm-resolvent rate on the outer arrow — is in the spirit of *multi-scale* or *two-parameter* convergence frameworks that appear elsewhere in NCG and mathematical physics:

- Semiclassical analysis (Sjöstrand–Vodev): combined semiclassical-parameter $\hbar \to 0$ and domain-size parameter $T \to \infty$ limits.
- Coarse Bloch decomposition: combined unit-cell-size and spectral-bandwidth parameters.
- Continuum limit of lattice gauge theories: combined lattice-spacing $a \to 0$ and lattice-volume $L \to \infty$ limits.

In each case, the two scales must be chosen jointly to give a well-defined limit. Our admissible scaling condition (Definition 2.6) is the analog in the Lorentzian-NCG setting.

## 8 Propinquity-level closure via temporal-Lipschitz-invisibility

*[Descoped (see the Status note): the construction below quantifies over the degenerate natural-substrate seminorm (whose kernel far exceeds the scalars; the temporal-Lipschitz-invisibility identity of Lemma 8.1 is the diagnosis of that degeneracy, not a feature). The section is retained as a record of the algebraic construction; its metric-level claims are open pending the upstream repair.]*

The norm-resolvent outer arrow of Theorem 5.1 can be upgraded to propinquity-level convergence on the non-compact carrier $S^3 \times \mathbb{R}_t$ by combining three ingredients: the temporal-Lipschitz-invisibility identity of Paper 46 (Lemma 3.2 [30]), the Latrémolière pinned proper QMS hypertopology [16], and its Krein-lift (Paper 48 [32] § 3). This *would* close Q1 (G2-metric) on the natural substrate, but the metric level is *descoped* (it inherits the Paper 45 degeneracy; see the Status note); only the norm-resolvent / spectral level (§§3–4) is established.

**Lemma 8.1** (Zero-cost extent element)**.** *Let $\chi_T \in C_c^\infty(\mathbb{R})$ be a smooth cutoff with $\chi_T \equiv 1$ on $[-(T-1)/2,\, (T-1)/2]$ and $\operatorname{supp}\chi_T \subseteq [-T/2,\, T/2]$. Define $e_T = \mathbf{1}_{S^3} \otimes \chi_T \in C_0(S^3 \times \mathbb{R})$. Then*

$$L^K(e_T) \;=\; \|[D_L,\, M_{e_T}]\|_{\mathrm{op}} \;=\; 0 \qquad \forall\, T > 0. \tag{23}$$

*Proof.* By the temporal-Lipschitz-invisibility identity (Paper 46 Lemma 3.2 [30]), every natural-substrate element $a = a_s \otimes a_t \in \mathcal{O}^L$ satisfies $[D_L, M_a] = i[D_{\mathrm{GV}}, M_{a_s}] \otimes M_{a_t}$. Setting $a_s = \mathbf{1}_{S^3}$ (the constant function on $S^3$) and $a_t = \chi_T$: $[D_L, M_{e_T}] = i[D_{\mathrm{GV}}, M_{\mathbf{1}}] \otimes M_{\chi_T} = 0$, since the Dirac operator commutes with the identity multiplier. □

**Remark 8.2.** Lemma 8.1 is the load-bearing step. In a general non-compact quantum metric space, the extent element's Lipschitz norm $L(e) > 0$ restricts the admissible metametric radius via the condition $L(e) \le r$ (Def. 2.6 of [16]). This forces $r$-dependent spatial localization—the source of the "multi-month" estimate in the original Q1 formulation. GeoVac's temporal-Lipschitz-invisibility gives $L^K(e_T) = 0 \le r$ for *every* $r > 0$: the temporal extent element is freely available at zero Lipschitz cost, and the non-compact extension adds nothing to the propinquity bound.

**Theorem 8.3** (G2-metric closure on the natural substrate)**.** *Let $\{(n_{\max}(k),\, N_t(k),\, T(k))\}_{k\in\mathbb{N}}$ be an admissible scaling sequence (Definition 2.6). Then the Krein pinned proper QMS sequence $\{\mathcal{T}^{L,\mathrm{pp}}_{n_{\max}(k),N_t(k),T(k)}\}$ converges in the Latrémolière hypertopology (Def. 4.4 of [16], Krein-lifted per Paper 48 [32] § 3) to $\mathcal{T}^{L,\mathrm{pp}}_{S^3\times\mathbb{R}_t}$:*

$$\eth^K_r\big(\mathcal{T}^L_{n_{\max}(k)},\, \mathcal{T}^L_{S^3\times\mathbb{R}}\big) \;\le\; C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma_{n_{\max}} \;+\; \epsilon(T(k)) \;\to\; 0 \qquad \forall\, r > 0, \tag{24}$$

*where $C_3^{\mathrm{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}}$ (Paper 46), $\gamma_{n_{\max}} = 2/(n_{\max}{+}1)$ (Paper 45 L2), and $\epsilon(T) \to 0$ as $T \to \infty$.*

*Proof. Step 1 (Continuum limit as Krein pinned proper QMS).* $\mathcal{T}^{L,\mathrm{pp}}_{S^3\times\mathbb{R}} = (\mathcal{A}^K,\, L^K,\, \mathcal{M}^L,\, \omega^L_W)$ with $\mathcal{A}^K = C_0(S^3 \times \mathbb{R})$, $L^K(f) = \|[D_L, M_f]\|_{\mathrm{op}}$, $\mathcal{M}^L = \operatorname{span}\{M^{\mathrm{spat}}_{N,L,0} \otimes I\}$, $\omega^L_W$ the wedge KMS state. The exhaustive sequence is $h_n = \mathbf{1} \otimes \chi_n$ with $L^K(h_n) = 0$ (Lemma 8.1), $\omega^L_W(h_n) \to 1$ (thermal tail decay), $\|h_n\| = 1$. All nine axioms of [16] Defs. 1.18, 1.22, 1.26, 1.29, 1.37, 1.40, 1.42 are verified at theorem-grade rigor in the Krein-lift (Paper 48 [32] § 3).

*Step 2 (M-tunnel construction).* For each $k$, construct an M-tunnel $\tau_k = (\mathfrak{D}_k,\, \pi_1,\, \pi_2,\, L_{\mathfrak{D}},\, e_k)$ with:

- $\mathfrak{D}_k = \mathcal{O}^L_{n_{\max}(k),N_t(k)} \oplus C_0(S^3 \times \mathbb{R})$ (bridging algebra);
- $\pi_1 = \mathrm{id}$, $\pi_2 = B^{\mathrm{SU}(2)}_{n_{\max}} \otimes (R_T \circ B^{S^1}_{N_t})$ (spatial Berezin ⊗ temporal restrict-Fejér);

- $L_{\mathfrak{D}}(a_1, a_2) = \max(L_1^K(a_1),\, L_2^K(a_2))$;
- $e_k = (I,\, \mathbf{1} \otimes \chi_{T(k)})$ with $L_{\mathfrak{D}}(e_k) = \max(0,\, 0) = 0$ (Lemma 8.1).

*Step 3 (Reach bound).* The tunnel reach decomposes into spatial and temporal contributions. The spatial reach is $\leq C_3^{\mathrm{op}} \cdot \gamma_{n_{\max}}$ by Paper 46 Theorem 5.1 (the compact-carrier propinquity bound, transported via the M-tunnel restriction to the extent region $|t| \leq T(k)/2$). The temporal contribution is controlled by the extent mechanism: for states $\phi$ with $\phi(e_k) \geq 1 - \delta$, the mass outside the extent region $[-T(k)/2,\, T(k)/2]$ is $\leq \delta$, giving a temporal reach $\leq \delta$. Setting $\delta = \epsilon(T(k))$ with $\epsilon(T) = 1 - \omega_W^L(\mathbf{1} \otimes \chi_T) \to 0$ (KMS thermal tail):

$$\chi(\tau_k,\, r) \;\leq\; C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma_{n_{\max}} \;+\; \epsilon(T(k)).$$

*Step 4 (Propinquity assembly).* For each $r > 0$: $L_{\mathfrak{D}}(e_k) = 0 \leq r$, so $\tau_k$ is admissible at radius $r$. By Def. 2.23 of [16] (Krein-lifted):

$$\eth_r^K(\mathcal{T}^L_{n_{\max}(k)},\, \mathcal{T}^L_{S^3\times\mathbb{R}}) \;\leq\; \chi(\tau_k,\, r) \;\leq\; C_3^{\mathrm{op}} \cdot \gamma_{n_{\max}} + \epsilon(T(k)) \;\to\; 0.$$

Since this holds for every $r > 0$, convergence in the hypertopology (Def. 4.4) follows. □

**Remark 8.4** (Bit-exact inheritance from the compact carrier)**.** As $T(k) \to \infty$, the temporal contribution $\epsilon(T(k)) \to 0$ and the bound is dominated by the compact-carrier term $C_3^{\mathrm{op}} \cdot \gamma_{n_{\max}}$. In particular, the Paper 46 panel values $\Lambda^L(2,3) = 2.0746$, $\Lambda^L(3,5) = 1.6101$, $\Lambda^L(4,7) = 1.3223$ are inherited as the asymptotic non-compact propinquity values. This is the "free upgrade" reading extended from the compact to the non-compact carrier.

**Remark 8.5** (Scope: natural substrate only)**.** Theorem 8.3 holds on the *natural substrate* (chirality-block-diagonal multipliers commuting with $J_L$, Paper 44 Prop. 5.1). On the *enlarged substrate* (Paper 46 Appendix B), the temporal-Lipschitz-invisibility identity fails: $[D_L^{\mathrm{diag}}, M^{\mathrm{flip}} \otimes M^{\mathrm{temp}}] \neq 0$ in general, so $L^K(e_T) > 0$ and the zero-cost extent mechanism does not apply. The enlarged-substrate non-compact extension remains open (Q2 below).

# 9 Open questions

The construction of §§ 3–4 closes G2 at the norm-resolvent level; the propinquity-level (hypertopology) upgrade of Theorem 8.3 is descoped (see the Status note), and the following open questions remain:

(Q1) *G2-metric: open (descoped 2026-06-09; see Status note).* (Theorem 8.3). The propinquity-level closure uses temporal-Lipschitz-invisibility (Paper 46 Lemma 3.2) to construct a zero-cost extent element, reducing the non-compact carrier to the compact one. The Paper 46 panel values are inherited bit-exactly.

(Q2) *Enlarged-substrate non-compact extension: open (descoped 2026-06-09; see Status note).* The $J_L$-graded gradient norm of Paper 46 Appendix B gives $G^{\mathrm{enlarged}}(f) = \|\nabla f^{\mathrm{block}}\|_\infty + 2\|\partial_t\|_{\mathrm{op}} \cdot \|f^{\mathrm{flip}}\|_\infty$. Under admissible scaling, $\|\partial_t\|_{\mathrm{op}} = O(N_t/T) \to \infty$ (since $T/N_t \to 0$). The enlarged Lip-1 constraint $G^{\mathrm{enlarged}} \leq 1$ forces $\|f^{\mathrm{flip}}\|_\infty \leq 1/(2\|\partial_t\|_{\mathrm{op}}) \to 0$: the chirality-flip content of the Lip-1 ball is *squeezed to zero* by the growing temporal Dirac norm. The enlarged-substrate metametric satisfies $\eth_r^{K,\mathrm{enlarged}} \leq C_3^{\mathrm{op}} \cdot \gamma_{n_{\max}} + 1/(2\|\partial_t\|_{\mathrm{op}}) + \epsilon(T) \to 0$, inheriting the Theorem 8.3 bound plus a vanishing flip-suppression correction. The $O(N_t/T)$ growth flagged in the original Q2 formulation is the convergence *mechanism*, not an obstruction.

(Q3) *Cross-manifold extension (G3): CLOSED as structural impossibility.* Paper 32 [31] Theorem 7.4[1] proves that the cross-manifold tensor product $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\mathrm{BL}}$ cannot preserve both the Riemannian-Dirac character of $S^3$ and the $\pi$-free Bargmann character of $S^5$ within the standard NCG framework. The obstruction is categorical (Riemannian vs complex-analytic), tracing to Bertrand's theorem combined with the Fock and HO rigidity theorems (Papers 23 [20], 24 [21]).

(Q4) *Inner-factor calibration (G4).* The Yukawa / Higgs selection question of the Connes–Marcolli almost-commutative extension is orthogonal to the convergence theorem and remains open.

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